Chapter 6 Breakout (Pt. II)

STAT 4714

Questions

1. A process yields 10% defective items. If 100 items are randomly selected from the process, what is the probability that the number of defectives
   (a) exceeds 13?
   (b) is less than 8?

$p = 0.10$, $n = 100$ $\Rightarrow b(x; 100, 0.1)$

$X = \#$ of defectives

(a) $P(X > 13) = 1 - P(X \leq 13)$

$= 1 - \text{pbinom}(13, 100, 0.10) = \underline{0.1239}$

$\approx 1 - P\left(Z \leq \frac{13.5 - 0.1(100)}{\sqrt{0.1(100)(0.9)}}\right)$

$= 1 - P(Z \leq 1.1667) = 1 - \text{pnorm}(1.1667, 0, 1)$

$= \underline{0.1217}$

(b) $P(X < 8) \approx P\left(Z \leq \frac{7.5 - 0.1(100)}{\sqrt{100(0.1)(0.9)}}\right)$

$= P(Z \leq -0.8333) = \text{pnorm}(-0.8333, 0, 1)$

$= \underline{0.202}$

2. The probability that a patient recovers from a delicate heart operation is 0.9. Of the next 100 patients having this operation, what is the probability that

   (a) between 84 and 95 inclusive survive?

   (b) fewer than 86 survive?

X = # of survived patients

$p = 0.9$, $n = 100$ $\Rightarrow$ $b(x; n, p)$

(a) $P(84 \le X \le 95) \approx P\left(\frac{83.5 - np}{\sqrt{np(1-p)}} < Z < \frac{95.5 - np}{\sqrt{np(1-p)}}\right)$

$= P(-2.1667 < Z < 1.8333)$

$=$ pnorm(1.8333, 0, 1) $-$ pnorm(-2.1667, 0, 1)

$= \underline{0.9515}$

(b) $P(X < 86) \approx P\left(Z \le \frac{85.5 - np}{\sqrt{np(1-p)}}\right)$

$= P(Z \le -1.5)$

$= \underline{0.0668}$

3. A pharmaceutical company knows that approximately 5% of its pills have an ingredient that is below the minimum strength, thus rendering the pill ineffective. What is the probability that fewer than 10 pills in a sample of 200 pills will be ineffective?

$p = 0.05$, $n = 200 \Rightarrow b(x; 200, 0.05)$

X = # of ineffective pills

$$P(X < 10) \approx P\left(Z \le \frac{9.5 - np}{\sqrt{np(1-p)}}\right)$$

$$= P(Z \le -0.1622)$$

$$= \text{pnorm}(-0.1622, 0, 1)$$

$$= \underline{0.4356}$$